

Naphthalene is the only golden benzenoid

Krystal Guo

Korteweg-de Vries Institute for Mathematics, University of Amsterdam

k.guo@uva.nl

Abstract

Benzene, naphthalene and triphenylene are the only polyhexes whose graphs have no eigenvalue in the open interval $(-\vartheta, \vartheta)$, where $\vartheta = (\sqrt{5} - 1)/2$ is the reciprocal of the golden ratio. In particular, naphthalene is the only polyhex of even order whose median eigenvalues are $\pm\vartheta$; this answers a question of Pisanski. The proof rests on a locality theorem: if A is the adjacency matrix of a polyhex with at least seven hexagons, then there is a connected set C of seven hexagons, one of which is the first hexagon of the polyhex in a fixed lexicographic order on the cells of the tiling, such that the principal submatrix of $A^2 - \vartheta^2 I$ on the vertices of C is not positive semidefinite. Seven is best possible. The locality theorem reduces to a finite computation, and we describe a certificate for it in which every spectral claim is an inequality between integers.

Keywords: chemical graph theory; benzenoid; median eigenvalues; HOMO–LUMO gap; golden ratio; spectral gap

MSC2020: 05C50 (primary); 05C92, 92E10, 15A18 (secondary).

1 Introduction

Take the hexagonal tiling of the plane, choose finitely many of its hexagons whose union is connected, and let X be the graph formed by the vertices and edges lying on the chosen hexagons. When the union has no holes, the chosen set of hexagons is a *benzenoid* and X is its graph: the carbon skeleton of a benzenoid hydrocarbon, with the hydrogen atoms suppressed, so that “carbon atom” and “vertex” are the same word here. Such a graph is planar, bipartite, has maximum degree three, and has no cycle of length less than six. The chemistry supplies the names and nothing else; names like naphthalene and coronene are labels for specific small graphs, and we will keep using them because the literature does.

In Hückel molecular orbital theory [14, 3, 13], the energy levels of the π -electrons of a conjugated hydrocarbon are, up to an affine change of units, the eigenvalues of the adjacency matrix of its carbon skeleton. A skeleton on n vertices carries n such electrons, and they occupy the $n/2$ largest eigenvalues, which are the $n/2$ lowest energies. Writing the eigenvalues as $\lambda_1 \geq \dots \geq \lambda_n$, the highest occupied molecular orbital corresponds to $\lambda_{n/2}$ and the lowest unoccupied one to $\lambda_{n/2+1}$, and their difference is the *HOMO–LUMO gap*, a basic index of kinetic stability. Mohar [16] proposed the study of these *median eigenvalues* as a problem in spectral graph theory in its own right, and showed in [17] that the median eigenvalues of a connected bipartite subcubic graph other than the Heawood graph lie in $[-1, 1]$; see also [18, 9]. For a bipartite graph, the pairing theorem of Coulson and Rushbrooke [2] says that the spectrum is symmetric about 0, so $\lambda_{n/2+1} = -\lambda_{n/2}$ and the HOMO–LUMO gap is $2\lambda_{n/2}$. The whole question is how close the innermost pair of eigenvalues comes to zero.

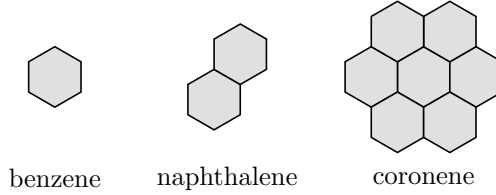


Figure 1: Three benzenoids, drawn as sets of cells of the hexagonal tiling. The graph is the union of the boundary cycles of the chosen cells; the hydrogen atoms are suppressed, so benzene is a 6-cycle, naphthalene has ten vertices and coronene twenty-four. Six of coronene’s vertices lie on three cells, and benzene and naphthalene have no such vertex; this distinction is the one that matters, and we will return to it in Section 2.

Which graphs keep an interval around zero free of eigenvalues is a spectral gap problem. Kollár and Sarnak [15] studied the intervals avoided by the spectra of all, or of infinitely many, cubic graphs, and the cubic graphs with no eigenvalue in $(-1, 1)$ and those with no eigenvalue in $(-2, 0)$ were characterized in [10, 11]; each of these two intervals is avoided by infinitely many cubic graphs. The question we answer here is of the same kind, for polyhexes and the interval $(-\vartheta, \vartheta)$, and the answer is finite.

Fowler and Pisanski [6] plotted the median eigenvalues of all connected graphs with maximum degree at most three on at most twelve vertices, and observed a conspicuous line of graphs with $\lambda_{n/2} = 1/\varphi$ exactly, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. Just over one percent of these graphs lie on the line, and they wrote that it is not clear why one eigenvalue should stand out in this way. Pisanski [21] called a graph *golden* when its median eigenvalues are exactly $\pm 1/\varphi$, listed the small golden graphs and the known golden fullerenes, amongst them the truncated icosahedron C_{60} [7], and asked whether naphthalene is the only golden benzenoid. In this paper we show that it is.

We will write

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \vartheta = \varphi^{-1} = \varphi - 1 = \frac{\sqrt{5} - 1}{2},$$

so $\varphi \approx 1.618034$ and $\vartheta \approx 0.618034$. Rather than asking which benzenoids attain ϑ , we ask which ones keep the open interval $(-\vartheta, \vartheta)$ free of eigenvalues. This is a formally stronger question, and its answer is the main result of the paper. We will state it for polyhexes, which are the connected unions of hexagons with holes permitted; benzenoids are the polyhexes without holes, so the result applies to them a fortiori. All terms are defined in Section 2.

Theorem 1.1 (golden gap theorem). *Let H be a polyhex. Then $X(H)$ has no eigenvalue in the open interval $(-\vartheta, \vartheta)$ if and only if H is benzene, naphthalene or triphenylene.*

The smallest positive eigenvalues of benzene, naphthalene and triphenylene are 1, ϑ and approximately 0.684040. Only naphthalene attains ϑ , and we obtain the answer to Pisanski’s question.

Corollary 1.2. *Naphthalene is the only golden benzenoid. More generally, it is the only polyhex of even order whose median eigenvalues are $\pm\vartheta$.*

Median eigenvalues are defined only for graphs of even order, and the restriction costs nothing: a polyhex of odd order has colour classes of different sizes, so 0 is an eigenvalue of its graph and it lies outside the golden gap class.

The proof of Theorem 1.1 has two parts. Polyhexes with at most six hexagons are finite in number and are checked directly. For the rest we prove a locality theorem, which says that the

failure to avoid $(-\vartheta, \vartheta)$ is always visible inside seven hexagons. To state it we need one piece of notation. For a set C of hexagons of a polyhex H with graph X and adjacency matrix A , let $V(C)$ be the set of vertices lying on the hexagons of C , and let $M_C(H)$ be the principal submatrix of $A^2 - \vartheta^2 I$ whose rows and columns are indexed by $V(C)$. We will see in Section 3 that X avoids $(-\vartheta, \vartheta)$ if and only if $A^2 - \vartheta^2 I$ is positive semidefinite, and that positive semidefiniteness is inherited by principal submatrices. Thus a single set C with $M_C(H)$ not positive semidefinite shows that X has an eigenvalue in $(-\vartheta, \vartheta)$; we will call such a set an obstruction. Cells of the tiling carry the coordinates fixed in Section 2 and are ordered lexicographically there. A translation of the tiling does not change the graph of a polyhex, so we may translate H so that its lexicographically first hexagon is the origin, and the seven hexagons we find can then be taken to include it.

Theorem 1.3 (seven-hexagon theorem). *Let H be a polyhex with at least seven hexagons, translated so that its lexicographically first hexagon is the origin. Then H contains a connected set C of exactly seven hexagons with $(0, 0) \in C$ such that $M_C(H)$ is not positive semidefinite.*

In particular, every polyhex with at least seven hexagons contains a connected seven-hexagon obstruction. The rooted form is the one the proof produces and the one the cross-checks of Section 7 test, and it turns the theorem into a search of bounded size around a single hexagon.

Seven cannot be replaced by six: we will exhibit a ten-hexagon benzenoid, not in the class of Theorem 1.1, in which every connected set of at most six hexagons has a positive semidefinite block (Proposition 4.2 and Corollary 4.3).

The proof of Theorem 1.3 is computer-assisted. The difficulty is that $M_C(H)$ depends not only on C but on the hexagons of H surrounding C , so the finite objects of the proof are not the 333 heptahexes, the connected sets of seven cells up to the symmetries of the tiling, but seven-hexagon cores decorated with partial information about their surroundings. We have made an effort to make the computation checkable. After a rooting reduction, the set of configurations that need to be examined is finite and explicit, and the proof is a certificate consisting of 3,652 rooted cores, 22,009 decorated patterns, each carrying an integer witness, and a covering directed acyclic graph with 345,554 nodes. Every spectral claim in the certificate is a strict inequality between integers, and the test is exact and not a relaxation, so no floating-point or algebraic-number comparison occurs on the decision path. The same is true of the other half of the proof of Theorem 1.1, the 116 polyhexes with at most six hexagons: 113 of them carry an integer witness decided by the same comparison, and the remaining three are benzene, naphthalene and triphenylene, whose characteristic polynomials are classical. The certificate has been checked by two independently written verifiers, and the resulting theorem has been cross-checked against exact spectra for all polyhexes with at most nine hexagons. Section 7 describes what was checked and what the computation does not establish.

The proof uses none of the structure theory of benzenoids. It does not use simple connectivity, Kekulé structures, the count of internal vertices, or the distinction between catacondensed and pericondensed systems. It is: a spectral reduction (Section 3), an explanation of why seven hexagons and their surroundings are the right objects together with the proof that seven is best possible (Section 4), the seven-hexagon theorem (Section 5), and the small cases (Section 6).

2 Polyhexes and benzenoids

The purpose of this section is to fix terminology; we defer to Gutman and Cyvin [12] for the theory of benzenoid systems and to Godsil and Royle [8] for graph-theoretic background.

Fix the infinite hexagonal tiling of the plane. Its bounded faces are *cells*; the chemical literature calls them *hexagons* or *rings*, and we will use the three words interchangeably. Every vertex of the tiling lies on exactly three cells, and each two of these three cells share an edge; thus two cells that share a vertex share an edge. Two distinct cells therefore either share exactly one edge, in which case we call them *adjacent*, or they are disjoint. A finite set of cells has a connected union if and only if the graph on the set whose edges are the adjacent pairs is connected. We will call this graph the *cell adjacency graph* of the set.

A *polyhex* is a finite nonempty set H of cells whose union is connected. Its *graph* $X(H)$ has as vertices the vertices of the tiling lying on cells of H , and as edges the edges of the tiling lying on cells of H ; in other words, $X(H)$ is the union of the boundary cycles of the cells of H . The graph $X(H)$ has cycles of length six of its own; we reserve the word *cycle* for cycles of $X(H)$ and never call them hexagons. We will write $h = |H|$ for the number of cells. A *benzenoid* is a polyhex whose union is simply connected, that is, whose union has no holes; a polyhex with holes is called a *coronoid* [4]. The smallest coronoid is the ring of six cells surrounding an empty cell, with $h = 6$. We will not need the distinction, since our results hold for all polyhexes. There are 1, 1, 3, 7, 22, 81, 331, 1435, 6505 benzenoids with $h = 1, \dots, 9$ cells [1], and 1, 1, 3, 7, 22, 82, 333, 1448, 6572 polyhexes [20], counted up to the symmetries of the tiling.

The graph $X(H)$ is planar, and since the tiling is bipartite, so is $X(H)$. Every vertex of $X(H)$ lies on one, two or three cells of H ; a vertex on one cell has degree 2, and a vertex on two or three cells has degree 3. A vertex lying on three cells of H is an *internal vertex*. A benzenoid is *catacondensed* if it has no internal vertices and *pericondensed* otherwise. The cell adjacency graph of a catacondensed benzenoid is a tree, and a catacondensed benzenoid has $4h + 2$ vertices; in general a benzenoid with n_i internal vertices has $4h + 2 - n_i$ vertices [12]. Every cycle of $X(H)$ has length at least six. In particular, $X(H)$ has no cycle of length four, and so two distinct vertices of $X(H)$ have at most one common neighbour.

Figure 2 shows five small benzenoids. Anthracene and phenanthrene both have three cells and 14 vertices, and differ only in whether the third cell is attached in line with the first two or at an angle. That difference already changes the spectrum: their smallest positive eigenvalues are approximately 0.4142 and 0.6052.

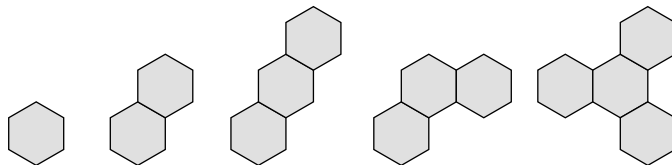


Figure 2: From left to right: benzene, naphthalene, anthracene, phenanthrene and triphenylene, drawn as sets of cells of the hexagonal tiling. Their smallest positive eigenvalues are 1, $\vartheta \approx 0.6180$, 0.4142, 0.6052 and 0.6840. The first two and the last are the members of the golden gap class.

We will need coordinates for cells. Use *axial coordinates*, in which the cells of the tiling are the pairs $(q, r) \in \mathbb{Z}^2$, and two cells are adjacent when they differ by one of the six vectors

$$(1, 0), \quad (-1, 0), \quad (0, 1), \quad (0, -1), \quad (1, -1), \quad (-1, 1).$$

The distance between cells in the cell adjacency graph of the whole tiling is then

$$d((q, r), (q', r')) = \max\{|q - q'|, |r - r'|, |(q + r) - (q' + r')|\},$$

and we write $d(q, r)$ for the distance from (q, r) to the origin. We order cells lexicographically by (q, r) . A translation of the tiling carries polyhexes to polyhexes with isomorphic graphs, so we may always translate a polyhex so that its lexicographically first cell is the origin.

3 Spectra and the golden gap

Let X be a graph with vertex set V . The *adjacency matrix* $A = A(X)$ is the symmetric 01-matrix with rows and columns indexed by V whose (u, v) -entry is 1 if and only if u and v are adjacent. We will use functional notation for the entries of matrices and vectors; the uv -entry of a matrix M is written $M(u, v)$, and the entry of a vector $\mathbf{y}$ indexed by u is written $\mathbf{y}(u)$. The *spectrum* $\text{spec } X$ of X is the multiset of eigenvalues of A , which are real since A is symmetric, and we will call these the *eigenvalues of X* . Order them $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$, where $n = |V|$. When n is even, the *median eigenvalues* of X are $\lambda_{n/2}$ and $\lambda_{n/2+1}$, and the *HOMO–LUMO gap* is their difference.

If X is bipartite, then λ and $-\lambda$ are eigenvalues of X with the same multiplicity, for every real λ ; this is the pairing theorem of Coulson and Rushbrooke [2]. Thus for a bipartite graph on an even number n of vertices the median eigenvalues are $\pm\lambda_{n/2}$, and they are $\pm\vartheta$ exactly when $\lambda_{n/2} = \vartheta$. Following Pisanski [21], we will say that a bipartite graph on an even number of vertices is *golden* if its median eigenvalues are $\pm\vartheta$. We use the word in Pisanski’s sense throughout; Estrada [5] uses the same phrase for a different class of graphs, defined by a condition on a ratio of spectral quantities, and Pisanski’s disambiguating name for the graphs considered here is *golden HOMO–LUMO graph*. A bipartite graph whose two colour classes have different sizes has 0 as an eigenvalue, with multiplicity at least the difference of the sizes, and so is not golden.

We will say that a polyhex H lies in the *golden gap class* if $X(H)$ has no eigenvalue in the open interval $(-\vartheta, \vartheta)$. A golden polyhex lies in the golden gap class, since its eigenvalues other than the median pair have absolute value at least ϑ by the pairing theorem. Conversely, a member of the golden gap class has no eigenvalue 0, so its order n is even, exactly $n/2$ of its eigenvalues are positive, and $\lambda_{n/2}$ is the smallest of these; and so it is golden if and only if ϑ is one of its eigenvalues. Thus Corollary 1.2 follows from Theorem 1.1 once we know the spectra of the three graphs in Figure 2; benzene is the 6-cycle, with spectrum $\pm 2, \pm 1, \pm 1$, naphthalene has ϑ as a simple eigenvalue, and the smallest positive eigenvalue of triphenylene is the root $0.684040\dots$ of $x^6 - 6x^4 + 9x^2 - 3$, a factor of its characteristic polynomial; these are readily computed.

3.1 Positive semidefiniteness

A real symmetric matrix M indexed by a set V is *positive semidefinite*, written $M \succeq 0$, if $\mathbf{y}^\top M \mathbf{y} \geq 0$ for every real vector $\mathbf{y}$ indexed by V ; equivalently, if every eigenvalue of M is nonnegative. For a subset $U \subseteq V$, the *principal submatrix* $M[U]$ is the matrix obtained from M by keeping the rows and columns indexed by U . If $\mathbf{y}$ is a vector indexed by U and $\tilde{\mathbf{y}}$ is its extension by zero to V , then $\mathbf{y}^\top M[U] \mathbf{y} = \tilde{\mathbf{y}}^\top M \tilde{\mathbf{y}}$. We obtain the following two facts, which we will use repeatedly and without further comment.

Lemma 3.1. *Let M be a real symmetric matrix indexed by V and let $U \subseteq V$.*

(a) *If $M \succeq 0$ then $M[U] \succeq 0$.*

(b) *If there is a vector $\mathbf{y}$ indexed by U with $\mathbf{y}^\top M[U] \mathbf{y} < 0$, then $M \not\succeq 0$.* □

We will call a vector $\mathbf{y}$ as in (b) a *witness* that M is not positive semidefinite. The proof of Theorem 1.3 consists entirely of producing witnesses; the small cases of Section 6 require the opposite, a certificate of positive semidefiniteness.

3.2 The spectral reduction

Since the eigenvalues of A^2 are the squares of the eigenvalues of A , the golden gap condition is a positive semidefiniteness condition.

Lemma 3.2. *Let X be a graph with adjacency matrix A . Then X has no eigenvalue in $(-\vartheta, \vartheta)$ if and only if*

$$A^2 - \vartheta^2 I \succeq 0.$$

If X is bipartite with colour classes S and T , and R is the submatrix of A with rows indexed by S and columns indexed by T , then this holds if and only if

$$RR^\top - \vartheta^2 I \succeq 0 \quad \text{and} \quad R^\top R - \vartheta^2 I \succeq 0.$$

Proof. The eigenvalues of $A^2 - \vartheta^2 I$ are the numbers $\lambda^2 - \vartheta^2$ for $\lambda \in \text{spec } X$, and $\lambda^2 \geq \vartheta^2$ if and only if $|\lambda| \geq \vartheta$. If X is bipartite then, ordering the vertices of S before those of T ,

$$A = \begin{pmatrix} 0 & R \\ R^\top & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} RR^\top & 0 \\ 0 & R^\top R \end{pmatrix},$$

and a block diagonal matrix is positive semidefinite if and only if each block is. $\square$

The matrix R is the *biadjacency matrix* of X , and the positive eigenvalues of X are the singular values of R ; when $|S| \neq |T|$ the matrix R is not square and 0 is an eigenvalue of X , as noted above.

The entries of A^2 are explicit: $A^2(u, u)$ is the degree of u , and for $u \neq v$ the entry $A^2(u, v)$ is the number of common neighbours of u and v . For the graph of a polyhex, the degrees are 2 or 3 and the number of common neighbours is 0 or 1, as we saw in Section 2. The following reformulation is not needed for the proof but explains where the golden ratio comes from. Since $\vartheta^2 = \vartheta \cdot \vartheta = (\varphi - 1)^2 = \varphi^2 - 2\varphi + 1 = 2 - \varphi$, we have $2 - \vartheta^2 = \varphi$.

Proposition 3.3. *Let H be a polyhex with graph X and colour classes S and T . Let Γ_S be the graph on S in which two vertices are adjacent when they have a common neighbour in X , and let D_S be the diagonal 01-matrix indexed by S with $D_S(u, u) = 1$ exactly when u has degree 3. We will call D_S the charge on S , and we will say that a vertex is charged when it has degree three in X . Then*

$$RR^\top - \vartheta^2 I = A(\Gamma_S) + D_S + \varphi I,$$

and H lies in the golden gap class if and only if the least eigenvalue of $A(\Gamma_S) + D_S$ and that of $A(\Gamma_T) + D_T$ are both at least $-\varphi$.

Proof. By the description of the entries of A^2 , we have $RR^\top = 2I + D_S + A(\Gamma_S)$, and $2 - \vartheta^2 = \varphi$. The second statement is Lemma 3.2. $\square$

One colour class of the tiling is a triangular lattice, in which two vertices are adjacent when they have a common neighbour in the tiling. The graph Γ_S is a subgraph of this lattice; it spans the subgraph induced on S , but in general it is a proper subgraph of that induced subgraph, since the common neighbour in the tiling of two adjacent vertices of S need not be a vertex of X at all. We will call a pair (Y, D) , consisting of a subgraph Y of the triangular lattice and a diagonal 01-matrix D indexed by the vertices of Y , a *charged subgraph* of the lattice, and we will call the least eigenvalue of $A(Y) + D$ the least eigenvalue of the pair. Thus Proposition 3.3 says that the golden gap class consists of those polyhexes for which two charged subgraphs of the triangular lattice have least eigenvalue at least $-\varphi$. For naphthalene, the least eigenvalue is exactly $-\varphi$; naphthalene is the equality case.

3.3 Local blocks

Let H be a polyhex with graph X and adjacency matrix A . For a set C of cells of H , we write $V(C)$ for the set of vertices of X lying on the cells of C , and we define

$$M_C(H) := (A^2 - \vartheta^2 I)[V(C)]. \quad (1)$$

Since X is bipartite, A^2 has no nonzero entry between the two colour classes, so every principal submatrix of $A^2 - \vartheta^2 I$ is block diagonal, with one block for the vertices of each colour class in its index set. We will call the two blocks of $M_C(H)$ its *colour blocks*; the matrix $M_C(H)$ is positive semidefinite if and only if both of its colour blocks are. We will say that C is an *obstruction* in H if $M_C(H) \not\geq 0$. By Lemmas 3.1 and 3.2, a polyhex containing an obstruction has an eigenvalue in $(-\vartheta, \vartheta)$. Note that $M_C(H)$ depends on all of H and not only on C , since the degrees of the vertices of $V(C)$ and their common neighbours are counted in $X(H)$. Note also that if $C' \subseteq C$ then $M_{C'}(H)$ is a principal submatrix of $M_C(H)$, so if C is not an obstruction then neither is any subset of C .

The dependence of $M_C(H)$ on H reaches no further than one ring of cells around C . The *collar* of C is the set of cells of the tiling that are not in C and are adjacent to a cell of C . The collar is a set of cells of the tiling, not of H ; some of its cells may lie in H and some may not.

Lemma 3.4. *Let C be a set of cells of a polyhex H . Then $M_C(H)$ is determined by C together with the set of cells of the collar of C that lie in H .*

Proof. The entries of $M_C(H)$ are the degrees of the vertices in $V(C)$ and the numbers of common neighbours of pairs of vertices in $V(C)$, and these are determined by the set of edges of $X(H)$ incident with $V(C)$. Every such edge lies on a cell of H containing a vertex $v \in V(C)$. Let c be such a cell and suppose $c \notin C$. Choose $c' \in C$ with v on c' . Exactly three cells of the tiling contain v , and they are pairwise adjacent; since c and c' are two of them, c is adjacent to c' and so lies in the collar of C . $\square$

4 Why the collar, and why seven

This section contains the motivation for the objects that the proof of Theorem 1.3 works with, and the proof that seven is best possible.

4.1 The block depends on the surroundings

Let C be a set of cells of a polyhex H . By Lemma 3.4, the block $M_C(H)$ is determined by C together with the cells of the collar of C that lie in H , and it does in general depend on those cells. Cells of H in the collar raise degrees and create common neighbours, and they may also add edges between vertices of $V(C)$. A set of cells that is an obstruction in isolation need not remain one once it is embedded in a larger polyhex. This is not a theoretical worry.

Example 4.1. Let C be the seven cells of coronene, a central cell with its six neighbours, and let $H = C$. Then each of the two colour blocks of $M_C(H)$ has least eigenvalue approximately -0.0912 , so C is an obstruction in coronene. Now let H be circumcoronene, which consists of C together with all twelve cells of its collar. Each colour block of $M_C(H)$ now has least eigenvalue approximately $+0.4100$; the block is positive definite, and C is not an obstruction in circumcoronene. (The two blocks have the same spectrum because a rotation of the tiling by 60° about the central cell exchanges the two colour classes.) Circumcoronene is not in the golden gap class, since it has

54 vertices and its smallest positive eigenvalue is approximately 0.3420; the obstruction is simply elsewhere.

Thus listing the 333 heptahexes is not enough. The finite objects in the proof of Theorem 1.3 are *decorated* sets of seven cells: a connected set C of seven cells, a partial assignment of the cells of its collar as present or absent, and a witness whose negativity survives every way of completing the assignment. We will make this precise in Section 5.

4.2 Seven is best possible

Theorem 1.3 uses seven cells, and seven is the least number for which such a statement is true. If a polyhex H is not in the golden gap class then the set of all of its cells is an obstruction; let $k(H)$ be the least k such that some connected set of k cells of H is an obstruction. Two small benzenoids settle all of $k = 1, \dots, 6$ between them. (For small k there is nothing to do: a polyhex in the golden gap class has no obstruction at all, so triphenylene already rules out $k \leq 4$.)

Proposition 4.2 (seven is best possible). *Let H_{10} be the benzenoid with cells*

$$(0, 3), (1, 1), (1, 2), (1, 4), (2, 2), (2, 3), (3, 0), (3, 1), (3, 3), (4, 1)$$

and let H_7 be the benzenoid with cells

$$(0, 0), (0, 3), (1, 0), (1, 1), (1, 2), (2, -1), (2, 2),$$

shown on the left and on the right of Figure 3. Neither lies in the golden gap class, and

$$k(H_{10}) = 7, \quad k(H_7) = 6.$$

Proof. The benzenoid H_{10} is catacondensed, with 42 vertices, and its smallest positive eigenvalue is $0.5311319792 \dots < \vartheta$; thus it is not in the golden gap class. It has

$$10, \quad 9, \quad 12, \quad 16, \quad 24, \quad 27$$

connected sets of cells of sizes $1, \dots, 6$, and symmetric Gaussian elimination, pivoting on positive diagonal entries and carried out exactly over $\mathbb{Q}(\sqrt{5})$ as described in Section 6, shows that every one of these sets C has $M_C(H_{10}) \geq 0$. Of the 23 connected sets of seven cells, 21 are obstructions, and so $k(H_{10}) = 7$.

The benzenoid H_7 is catacondensed as well. It has six connected sets of five cells, none of which is an obstruction, while all four of its connected sets of six cells are. Since H_7 is connected and has seven cells, every connected set of at most four of its cells extends to a connected set of five cells; a subset of a set that is not an obstruction is not an obstruction, so no connected set of at most five cells of H_7 is an obstruction and $k(H_7) = 6$. In particular H_7 contains an obstruction, so it is not in the golden gap class either. $\square$

Corollary 4.3. *For no $k \leq 6$ does every polyhex with at least k cells contain a connected set of k cells that is an obstruction.*

Proof. Let $k \leq 6$. If $k = 6$, then H_{10} has at least k cells and, since $k(H_{10}) = 7$, no connected set of k cells of H_{10} is an obstruction. If $k \leq 5$, then H_7 has at least k cells and, since $k(H_7) = 6$, no connected set of k cells of H_7 is an obstruction. $\square$

The smaller witness already settles $k \leq 5$, so the ten cells of H_{10} are needed only for $k = 6$. Note the mechanism. The obstruction can stay invisible at scale six even in a benzenoid that is comfortably outside the golden gap class, so the seven in Theorem 1.3 is a feature of the problem and not an artefact of the method.

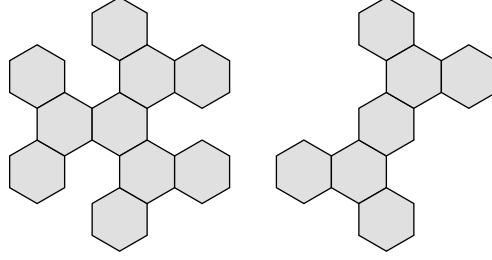


Figure 3: The two witnesses of Proposition 4.2. Left: the catacondensed benzenoid H_{10} , in which no connected set of at most six cells is an obstruction, although 21 of its 23 connected seven-cell sets are. Right: the seven-cell benzenoid H_7 , in which no connected set of at most five cells is an obstruction, although all four of its connected six-cell sets are.

5 Seven hexagons suffice

This section proves Theorem 1.3. The reduction to a finite problem is elementary; the finite problem is then discharged by a certificate whose verification we describe in Section 7.

5.1 Rooting

Let H be a polyhex with at least seven cells, translated so that its lexicographically first cell is the origin $(0, 0)$. Every cell of H then lies in the half-plane

$$\Lambda^+ = \{(q, r) : q > 0\} \cup \{(0, r) : r \geq 0\}. \quad (2)$$

This is immediate from the definition of the lexicographic order on cells. We will call a polyhex whose lexicographically first cell is the origin a *rooted polyhex*; every rooted polyhex is contained in Λ^+ , and that containment is what the arguments below use. For $j \geq 0$, put $W_j = \{(q, r) \in \Lambda^+ : d(q, r) \leq j\}$. The map $(q, r) \mapsto (-q, -r)$ pairs the cells at distance at most j from the origin, other than the origin itself, and Λ^+ contains exactly one cell of each pair; since there are $3j^2 + 3j + 1$ cells at distance at most j , it is easy to see that $|W_j| = (3j^2 + 3j + 2)/2$. In particular

$$|W_6| = 64, \quad |W_7| = 85. \quad (3)$$

A *rooted core* is a connected set of seven cells of Λ^+ that contains the origin. Since a rooted core is connected and contains the origin, each of its cells is within distance six of the origin, so every rooted core lies in W_6 . Expanding connected sets one cell at a time inside Λ^+ , we find that there are exactly

$$3,652 \quad (4)$$

rooted cores. This is a rooted count: once the half-plane has been fixed, no rotation or reflection is factored out. The polyhex H contains a rooted core: run a breadth-first search in the cell adjacency graph of H from the origin and stop when seven cells have been found.

Lemma 5.1. *Let H be a rooted polyhex and let C be a rooted core contained in H . Then every cell of the collar of C that lies in H lies in W_7 . Thus $M_C(H)$ is determined by C and by the set $H \cap W_7$.*

Proof. A cell of the collar of C is adjacent to a cell of $C \subseteq W_6$, so it is within distance seven of the origin; if it lies in H then it lies in Λ^+ by (2), and so in W_7 . The second statement now follows from Lemma 3.4. $\square$

We will write $\text{collar}^+(C)$ for the set of cells of the collar of C that lie in Λ^+ , and we will call it the *admissible collar* of C . By Lemma 5.1 these are exactly the collar cells that can lie in a rooted polyhex, and they lie in W_7 ; a collar cell outside Λ^+ is absent from every rooted polyhex, and we will treat it throughout as forced absent.

We will call the set W_7 the *window*, and we will call a subset of W_7 an *occupancy pattern*. By Lemma 5.1, once H is rooted, the block $M_C(H)$ for any rooted core $C \subseteq H$ is a function of the occupancy pattern $H \cap W_7$, an 85-bit object. There are finitely many occupancy patterns, and Theorem 1.3 has become a finite statement. Figure 4 shows the window with a rooted core and its collar. The core drawn there is coronene, a central cell with its six neighbours, placed so that its lexicographically first cell is the origin; its collar has twelve cells, of which eight lie in W_7 and four lie outside Λ^+ and are therefore absent from every rooted polyhex. With all eight permitted collar cells present, the two colour blocks of M_C have least eigenvalues approximately $+0.0141$ and $+0.1998$, so this core is not an obstruction in that configuration, and the cover of Section 5.3 must supply a different core.

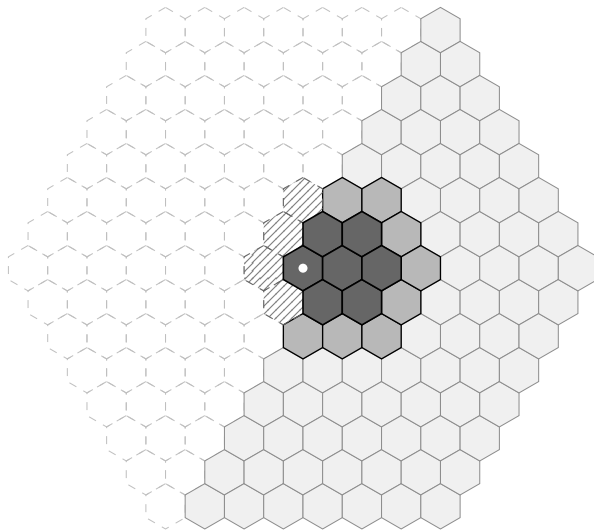


Figure 4: The window W_7 (the 85 solid cells), with the origin marked by a white dot. The dashed cells lie outside the half-plane Λ^+ and are absent from every rooted polyhex. The dark cells are the rooted core consisting of the seven cells of coronene, placed so that its lexicographically first cell is the origin. Of the twelve cells of its collar, the eight lying in W_7 are medium grey and the four lying outside Λ^+ are hatched.

5.2 Decorated patterns and integer certificates

The tiling is bipartite; we fix one of its two colour classes and call its vertices *black* and the others *white*. The graph of every polyhex inherits this colouring. Every edge of the tiling lies on exactly two cells, and it is an edge of $X(H)$ if and only if at least one of these two cells lies in H .

A *decorated pattern* is a quadruple

$$P = (C; P^+, P^-; \mathbf{y}),$$

where C is a rooted core, P^+ and P^- are disjoint subsets of W_7 consisting of cells of C or of its collar, with $C \subseteq P^+$, and $\mathbf{y}$ is a nonzero integer vector indexed by the vertices lying on the cells of

C , whose support consists of vertices of a single colour. We will refer to the cells of P^+ as *forced present* and to the cells of P^- as *forced absent*. An occupancy pattern $\Omega \subseteq W_7$ matches P if $P^+ \subseteq \Omega$ and $P^- \cap \Omega = \emptyset$, and a rooted polyhex H matches P if $H \cap W_7$ does. Note that $C \subseteq H$ whenever H matches P .

Fix a decorated pattern P and suppose that $\mathbf{y}$ is supported on black vertices. Let w be a white vertex of the tiling joined by an edge e of the tiling to a vertex of the support of $\mathbf{y}$. The status of e in a polyhex matching P is one of three kinds. If one of the two cells on e lies in P^+ , then e is an edge of $X(H)$ for every H matching P , and we say that e is *forced*. If every cell on e lies in P^- or outside Λ^+ , equivalently if no cell on e lies in $C \cup \text{collar}^+(C)$ outside P^- , then e is an edge of no such $X(H)$ and we call it *discarded*. Otherwise e is *undetermined*. Let s_w be the sum of the entries $\mathbf{y}(v)$ over the forced edges vw at w , and let $t_{w,1}, \dots, t_{w,k_w}$ be the entries $\mathbf{y}(v)$ over the undetermined edges vw at w ; since w has degree three in the tiling, $k_w \leq 3$. Define

$$\hat{q}_P(\mathbf{y}) = \sum_w \max_{J \subseteq \{1, \dots, k_w\}} \left(s_w + \sum_{j \in J} t_{w,j} \right)^2 - 2\mathbf{y}^\top \mathbf{y}, \quad (5)$$

the sum running over all white vertices w as above. This is an integer. If instead $\mathbf{y}$ is supported on white vertices, we define $\hat{q}_P(\mathbf{y})$ by the same formula with the two colours exchanged, and everything below applies verbatim.

Lemma 5.2. *Let $P = (C; P^+, P^-; \mathbf{y})$ be a decorated pattern. For every rooted polyhex H matching P , with adjacency matrix A ,*

$$\mathbf{y}^\top (A^2 - 2I) \mathbf{y} \leq \hat{q}_P(\mathbf{y}).$$

Proof. Regard $\mathbf{y}$ as a vector indexed by the vertices of $X(H)$, extended by zero. We may assume that $\mathbf{y}$ is supported on black vertices, the other case being the same with the two colours exchanged. Since $X(H)$ is bipartite, the black coordinates of $A\mathbf{y}$ vanish, and so

$$\mathbf{y}^\top A^2 \mathbf{y} = \|A\mathbf{y}\|^2 = \sum_w \left(\sum_{v \sim w} \mathbf{y}(v) \right)^2,$$

where the outer sum runs over the white vertices w of $X(H)$ and the inner sum over the neighbours v of w in $X(H)$. Only white vertices joined to the support of $\mathbf{y}$ contribute, and for such a vertex w the inner sum is $s_w + \sum_{j \in J_w} t_{w,j}$, where J_w indexes the undetermined edges at w that are edges of $X(H)$. Replacing each J_w by the subset that maximizes the square can only increase the total. The sum in (5) runs over the white vertices of the tiling joined to the support, and these need not all be vertices of $X(H)$; a white vertex of the tiling that is not a vertex of $X(H)$ contributes nothing to $\|A\mathbf{y}\|^2$ and a nonnegative amount to $\hat{q}_P(\mathbf{y})$, so the inequality is unaffected. Subtracting $2\mathbf{y}^\top \mathbf{y}$ from both sides gives the result. $\square$

The point of the relaxation is that the maximum at each w is taken separately, ignoring the constraint that the sets J_w at different vertices arise from one common set of cells. This makes $\hat{q}_P(\mathbf{y})$ computable by inspecting at most 2^3 subsets at each of a handful of vertices, and it is sharp enough for our purposes.

The condition we need is $\hat{q}_P(\mathbf{y}) + \varphi \mathbf{y}^\top \mathbf{y} < 0$, and although φ is irrational this is an inequality between integers. Write

$$N = \mathbf{y}^\top \mathbf{y}, \quad M = -\hat{q}_P(\mathbf{y}),$$

both integers, with $N > 0$ since $\mathbf{y} \neq 0$. Then $\hat{q}_P(\mathbf{y}) + \varphi N < 0$ says exactly that $\varphi N < M$, and since $N > 0$,

$$\varphi N < M \iff \sqrt{5} N < 2M - N \iff 2M - N > 0 \text{ and } 5N^2 < (2M - N)^2, \quad (6)$$

the last step because the left side of the middle inequality is positive, so squaring is reversible exactly when the right side is positive too. We will say that P is a *robust obstruction pattern* if the two strict integer inequalities on the right of (6) hold. No algebraic number and no floating point quantity occurs in that test.

Lemma 5.3. *Let $P = (C; P^+, P^-; \mathbf{y})$ be a robust obstruction pattern. Then for every rooted polyhex H matching P we have $\mathbf{y}^\top M_C(H) \mathbf{y} < 0$. In particular, C is an obstruction in H .*

Proof. Since $\mathbf{y}$ is supported on $V(C)$, we have $\mathbf{y}^\top M_C(H) \mathbf{y} = \mathbf{y}^\top (A^2 - \vartheta^2 I) \mathbf{y}$. Using $\vartheta^2 = 2 - \varphi$, then Lemma 5.2, and finally (6),

$$\mathbf{y}^\top (A^2 - \vartheta^2 I) \mathbf{y} = \mathbf{y}^\top (A^2 - 2I) \mathbf{y} + \varphi \mathbf{y}^\top \mathbf{y} \leq \hat{q}_P(\mathbf{y}) + \varphi \mathbf{y}^\top \mathbf{y} < 0. \quad \square$$

Example 5.4. The simplest rooted core is the straight chain of seven cells $(0, 0), (0, 1), \dots, (0, 6)$, the heptacene, rooted at one end and shown in Figure 5. The cells $(-1, r)$ for $r = 0, \dots, 7$ to its left and the cell $(0, -1)$ below it lie outside Λ^+ and are absent from every rooted polyhex; they are drawn dashed in Figure 5. The left side of the chain is a path on which black and white vertices alternate. Each black vertex on it lies on one cell of C and on two of the absent cells, and so has degree two. Each white vertex on it, apart from the bottom one, which also has degree two, is the left endpoint of one of the six *fusion edges* of the chain, the edges shared by two adjacent cells, and has degree three. Take $P^+ = C$, $P^- = \emptyset$, and let $\mathbf{y}$ take the values $-2, 4, -5, 5, -4$ on the five black vertices of the left side of the first five cells, and 0 elsewhere. The support has eleven white tiling-neighbours, and the only ones that contribute are the six on this path; the other five lie only on cells outside Λ^+ , so all of their edges are discarded and they contribute 0. At each of the six contributing vertices, every edge of the tiling either lies on a cell of C , namely the two edges of the boundary path and, at a vertex of degree three, the fusion edge, or lies on two cells outside Λ^+ ; thus there are no undetermined edges, and the values s_w are $-2, 2, -1, 0, 1, -4$. Thus

$$\mathbf{y}^\top \mathbf{y} = 4 + 16 + 25 + 25 + 16 = 86, \quad \hat{q}_P(\mathbf{y}) = (4 + 4 + 1 + 0 + 1 + 16) - 2 \cdot 86 = -146,$$

so $N = 86$ and $M = 146$, giving $2M - N = 206 > 0$ and $5N^2 = 36,980 < 42,436 = (2M - N)^2$. Since no edge at the support is undetermined, the bound of Lemma 5.2 is an equality here, and in fact $\mathbf{y}^\top M_C(H) \mathbf{y} = 86\varphi - 146 \approx -6.85$ for every rooted polyhex H containing this chain. The single pattern P thus covers every occupancy pattern containing this core, whatever is attached on the right of the chain.

Most patterns are less simple than Example 5.4. The witnesses of the certificate have between 7 and 15 nonzero entries, with 13 the commonest number, and most patterns need some cells of the collar to be forced present or forced absent before the integer inequality (6) holds; this is exactly why a cover is needed. The pattern of Example 5.4 is not one of the 22,009, and we give it because it is small enough to be checked by hand: the certificate is not claimed to be minimal, and shorter witnesses exist for many of its cores.

5.3 The cover

What follows is the standard first-failure branching of a backtracking search over the occupancy patterns, recorded as a static object so that it can be checked rather than re-run.

The occupancy patterns are the vertices of a Boolean cube of dimension 85, with one coordinate for each cell of the window. A *subcube* is a pair $\mathcal{Q} = (U^+, U^-)$ of disjoint subsets of W_7 , and an occupancy pattern Ω lies in $\mathcal{Q}$ if $U^+ \subseteq \Omega$ and $U^- \cap \Omega = \emptyset$. The cells of U^+ are the cells that $\mathcal{Q}$

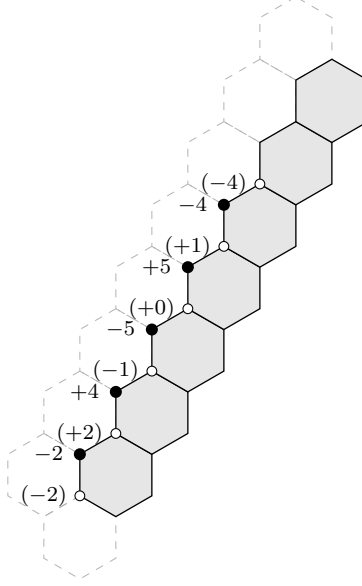


Figure 5: The worked example of Example 5.4. The heptacene core is rooted at its bottom cell; the dashed cells lie outside Λ^+ . Filled vertices carry the entries of the witness $\mathbf{y}$, and the hollow vertices are the white vertices w at which $\|\mathbf{A}\mathbf{y}\|^2$ is evaluated, labelled by the sums s_w in parentheses.

declares present and those of U^- are the cells it declares absent; the remaining cells of W_7 are the cells undecided by $\mathcal{Q}$. The *literals* of a decorated pattern $P = (C; P^+, P^-; \mathbf{y})$ are the statements “ c is present” for $c \in P^+$ and “ c is absent” for $c \in P^-$; an occupancy pattern matches P exactly when it satisfies all the literals of P . A literal of P is *decided* by a subcube $\mathcal{Q}$ if it is satisfied or violated by every occupancy pattern in $\mathcal{Q}$, and *undecided* otherwise; P is *consistent* with $\mathcal{Q}$ if none of its literals is violated by every occupancy pattern in $\mathcal{Q}$.

Suppose P is consistent with $\mathcal{Q}$ and let $\ell_1, \dots, \ell_m$ be the literals of P undecided by $\mathcal{Q}$, listed in a fixed order. Then the occupancy patterns in $\mathcal{Q}$ are partitioned into $m + 1$ classes: those satisfying all of $\ell_1, \dots, \ell_m$, which match P ; and, for each $i = 1, \dots, m$, those satisfying $\ell_1, \dots, \ell_{i-1}$ and violating ℓ_i . Each of the latter classes is again the set of occupancy patterns in a subcube, which we denote $\mathcal{Q}_i$; it is obtained from $\mathcal{Q}$ by deciding the cells of $\ell_1, \dots, \ell_i$, with ℓ_i decided the wrong way. We will call $\mathcal{Q}_1, \dots, \mathcal{Q}_m$ the *first-failure children* of $(P, \mathcal{Q})$.

The *cover* is a directed acyclic graph. Each node stores a robust obstruction pattern P and a subcube $\mathcal{Q}$ consistent with P , and its out-neighbours are nodes whose subcubes are exactly the first-failure children of $(P, \mathcal{Q})$, one for each child. A node is a *leaf* if P has no literal undecided by $\mathcal{Q}$. For each rooted core C_0 there is a *root* node whose subcube is $(C_0, \emptyset)$: the seven cells of C_0 are present and the remaining 78 cells of W_7 are undecided. Nodes may have several in-neighbours; subproblems that arise more than once are shared. The certificate consists of

$$3,652 \text{ rooted cores}, \quad 22,009 \text{ robust obstruction patterns}, \quad 345,554 \text{ cover nodes.} \quad (7)$$

Example 5.5. We give one node of the cover, taken from the certificate, together with its children; it is drawn in Figure 6. Let

$$C_0 = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0)\}$$

be a rooted core, and consider its root node, whose subcube is $\mathcal{Q} = (C_0, \emptyset)$: the seven cells of C_0 are present and the other 78 cells of W_7 are undecided. The pattern P stored at this node has

core C_0 , with $P^+ = C_0$ and $P^- = \{(1, -1), (2, -1)\}$, two cells of the collar of C_0 . Its witness has thirteen nonzero entries, and

$$\begin{aligned} N = \mathbf{y}^\top \mathbf{y} &= 104, & M = -\hat{q}_P(\mathbf{y}) &= 186, \\ 2M - N &= 268 > 0, & 5N^2 &= 54,080 < 71,824 = (2M - N)^2, \end{aligned}$$

so P is a robust obstruction pattern.

The pattern P has nine literals. The subcube decides the seven that assert that the cells of C_0 are present, and leaves two undecided: ℓ_1 , that $(1, -1)$ is absent, and ℓ_2 , that $(2, -1)$ is absent. The node therefore has two first-failure children. The first is the subcube in which $(1, -1)$ is present and nothing else has been decided beyond C_0 , and the second is the subcube in which $(1, -1)$ is absent and $(2, -1)$ is present. Both children are leaves. The pattern stored at the first has core

$$\{(0, 0), (0, 1), (0, 2), (1, -1), (1, 0), (1, 1), (1, 2)\},$$

which is C_0 with $(2, 0)$ traded for $(1, -1)$, and the pattern stored at the second has core

$$\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, -1)\},$$

which is C_0 with $(2, 0)$ traded for $(2, -1)$. Neither of these two patterns has any literal beyond the seven cells of its own core, so each of them is a leaf under every subcube containing its core, and three patterns between them handle every occupancy pattern containing C_0 . The two literals ℓ_1 and ℓ_2 are what the integer inequality (6) needs rather than what the geometry needs: C_0 is in fact an obstruction in the polyhex $C_0 \cup \{(1, -1)\}$ as well, but the relaxation $\hat{q}_P$ does not certify it there. In the certificate this is core 280 and node 599, storing pattern 513, with leaves 416 and 598 storing patterns 342 and 512.

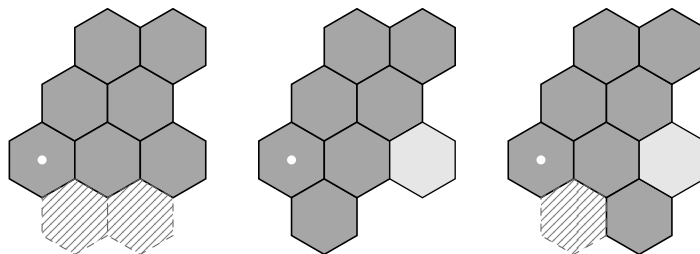


Figure 6: The node of Example 5.5 and its two first-failure children, with the origin marked by a white dot. In each drawing the core of the stored pattern is dark, a cell declared present by the subcube but not in that core is light, and a hatched cell is one declared absent; the cells of W_7 that are not drawn are undecided. Left: the root node for C_0 , whose pattern forces the two collar cells $(1, -1)$ and $(2, -1)$ absent. Centre: the child in which $(1, -1)$ is present. Right: the child in which $(1, -1)$ is absent and $(2, -1)$ is present. In each child the core that carries the witness is a different seven-cell set.

Lemma 5.6. *Let C_0 be a rooted core and let Ω be an occupancy pattern containing C_0 . Then Ω matches the pattern stored at some node of the cover.*

Proof. We show by induction on the number of undecided cells of the subcube that every occupancy pattern lying in the subcube of a node matches the pattern stored at that node or at some node reachable from it. At a leaf every literal of P is decided by $\mathcal{Q}$; since P is consistent with $\mathcal{Q}$, no

literal of P is violated by every occupancy pattern in $\mathcal{Q}$, and a decided literal is either satisfied by all of them or violated by all of them, so every literal of P is satisfied by every occupancy pattern in $\mathcal{Q}$. Thus at a leaf every occupancy pattern in the subcube matches the stored pattern. At a node $(P, \mathcal{Q})$ that is not a leaf, an occupancy pattern in $\mathcal{Q}$ either matches P , or has a first undecided literal of P that it violates and so lies in a first-failure child, which is the subcube of an out-neighbour and has strictly fewer undecided cells. Since Ω lies in the subcube $(C_0, \emptyset)$ of the root node for C_0 , the statement follows. $\square$

Proof of Theorem 1.3. Let H be a polyhex with at least seven cells, translated so that its lexicographically first cell is the origin. As observed in Section 5.1, H contains a rooted core C_0 , and so the occupancy pattern $\Omega = H \cap W_7$ contains C_0 . By Lemma 5.6, Ω matches some robust obstruction pattern $P = (C; P^+, P^-; \mathbf{y})$ of the cover. Then H matches P , and $C \subseteq P^+ \subseteq H$ is a connected set of exactly seven cells. The core of a decorated pattern is a rooted core, so $(0, 0) \in C$. By Lemma 5.3, $\mathbf{y}^\top M_C(H) \mathbf{y} < 0$, so C is an obstruction in H . $\square$

Remark 5.7. The core C supplied by the cover need not be the core C_0 we started from, and this freedom is essential rather than cosmetic. In the configuration of Figure 4, the coronene core is not an obstruction, and the cover certifies the configuration through a different seven-cell set inside the same 85-cell window.

Remark 5.8. The certificate is sufficient and has been verified exactly, but it is not claimed to be minimal, irredundant or in any sense canonical. Both the number of patterns and the sizes of their witnesses can be reduced, at the cost of a larger cover. We report the certificate as it stands because it is the one that has been verified.

6 Polyhexes with at most six hexagons

Generate all polyhexes with at most six cells, up to translations and the dihedral symmetries of the tiling, by adjoining one adjacent cell at a time and canonicalizing. The counts are

$$1, 1, 3, 7, 22, 82 \tag{8}$$

for $h = 1, \dots, 6$, a total of 116 polyhexes; the ring of six cells around a hole is amongst them. Exactly three of the 116 lie in the golden gap class, and the two cases call for opposite kinds of evidence: for the others we exhibit a witness, and for these three we must certify that none exists.

6.1 The hundred and thirteen that are not in the class

Let H be one of the 113 polyhexes that are not in the golden gap class, with graph X and adjacency matrix A , and let $\mathbf{y}$ be a nonzero integer vector supported on one colour class of X . Here H is completely known, so nothing plays the role that $\hat{q}_P$ played in Section 5.2: the quantity $\mathbf{y}^\top (A^2 - 2I) \mathbf{y}$ is computed exactly. Put

$$N = \mathbf{y}^\top \mathbf{y}, \quad M = -\mathbf{y}^\top (A^2 - 2I) \mathbf{y},$$

both integers, with $N > 0$. Since $\vartheta^2 = 2 - \varphi$, we have $\mathbf{y}^\top (A^2 - \vartheta^2 I) \mathbf{y} = \varphi N - M$, so $\mathbf{y}$ witnesses that $A^2 - \vartheta^2 I$ is not positive semidefinite precisely when $\varphi N < M$, which by (6) is the pair of integer inequalities

$$2M - N > 0 \quad \text{and} \quad 5N^2 < (2M - N)^2. \tag{9}$$

Such a $\mathbf{y}$ exists for each of the 113. Indeed A^2 is block diagonal with respect to the two colour classes, so one of the two blocks fails to be positive semidefinite, and a witness supported on a single colour class is obtained from an eigenvector of that block for a negative eigenvalue by rounding and scaling. We record one for each of the 113. They are small: each has at most 11 nonzero entries, no entry exceeds 4 in absolute value, and 69 of the 113 have every entry equal to ± 1 . The largest integer occurring in (9) anywhere among them is 12,544.

6.2 The three that are

For benzene, naphthalene and triphenylene we give the characteristic polynomial and its factorization over $\mathbb{Z}$. Writing χ for the characteristic polynomial,

$$\begin{aligned}\chi_{\text{benzene}} &= (x-2)(x+2)(x-1)^2(x+1)^2, \\ \chi_{\text{naphthalene}} &= (x-1)(x+1)(x^2-x-3)(x^2+x-3)(x^2-x-1)(x^2+x-1), \\ \chi_{\text{triphenylene}} &= (x^3-3x^2+3)(x^3+3x^2-3)(x^6-6x^4+9x^2-3)^2.\end{aligned}$$

The smallest positive root of χ_{benzene} is 1. Now ϑ is the positive root of x^2+x-1 , so ϑ is a root of $\chi_{\text{naphthalene}}$, and it is the smallest positive one; naphthalene is therefore in the golden gap class and attains ϑ . The smallest positive root of $\chi_{\text{triphenylene}}$ is the root $0.684040\dots > \vartheta$ of the sextic factor $x^6-6x^4+9x^2-3$. Each of these statements is a finite check on integer polynomials: it suffices to evaluate each factor at ϑ in $\mathbb{Z}[\varphi]$, where $\vartheta = \varphi - 1$, and to count roots in $(0, \vartheta]$.

Proposition 6.1. *Exactly three of the 116 polyhexes with at most six cells lie in the golden gap class: benzene, naphthalene, and the four-cell benzenoid with cells $(0, 1), (1, 1), (1, 2), (2, 0)$, which is triphenylene.*

Proof. For each of the other 113, the recorded witness satisfies (9), so by Lemma 3.2 its graph has an eigenvalue in $(-\vartheta, \vartheta)$. For the three named, the factorizations above show that no eigenvalue lies in that interval. $\square$

Every comparison in this section is therefore between integers, apart from the three characteristic polynomials, whose roots are classical. As an independent cross-check we also decide positive semidefiniteness of the two matrices $RR^\top - \vartheta^2 I$ and $R^\top R - \vartheta^2 I$ of Lemma 3.2 directly, for all 116 polyhexes, by symmetric Gaussian elimination pivoting on positive diagonal entries and carried out exactly over $\mathbb{Q}(\sqrt{5})$: each entry is stored as $(a + b\varphi)/c$ with arbitrary precision integers a, b, c , and the sign of each entry is decided exactly. At each step we inspect the diagonal of the block that remains. A negative diagonal entry certifies failure. A zero diagonal entry of a positive semidefinite matrix forces its row and its column to vanish, so if every remaining diagonal entry is zero we return positive semidefinite exactly when the whole remaining block is zero. Otherwise we pivot on a positive diagonal entry, which leaves a smaller symmetric block, and repeat. This decides positive semidefiniteness and not merely positive definiteness, and the distinction is exactly what is needed here: naphthalene is the equality case, so its blocks are singular and the elimination stalls on them. The two computations agree.

Proof of Theorem 1.1. Let H be a polyhex. If H has at least seven cells, then by Theorem 1.3 it contains an obstruction, and by Lemmas 3.1 and 3.2 its graph has an eigenvalue in $(-\vartheta, \vartheta)$. If H has at most six cells, then by Proposition 6.1 and Lemma 3.2 it lies in the golden gap class if and only if it is benzene, naphthalene or triphenylene. $\square$

7 Verification

We now describe what was computed, how it was checked, and what the computation does not establish. The certificate, the verifiers and the transcripts of their runs accompany this paper.¹

The certificate and its verifier

The certificate is a text file listing the 85 cells of W_7 , the 3,652 rooted cores, and the 22,009 robust obstruction patterns, together with a binary file encoding the 345,554 nodes of the cover. The two files and their SHA-256 digests are

```
seven_hexagon_obstructions.txt
37936464fe433b52987dad9f5c2f186de52fc6ef0850c976d778d0c9a755ab78
seven_hexagon_cover.bin
51183fa42b2592ebd276e793a7d9ba9b7bc7a0f8633db4a5fc536e4b177fbb6b
```

so that the transcripts quoted below are tied to exact bytes.

The two formats are specified in Appendix A, in enough detail to be read without reference to our code.

The certificate is checked by `seven_hexagon_exact_verify.cpp`, a C++ program of about 350 lines which does not trust the search that produced it. The program

- (i) reconstructs W_6 and W_7 and verifies the counts (3), and enumerates the rooted cores independently and compares them, in order, with those listed;
- (ii) for each pattern, re-derives the collar of its core, checks that P^+ and P^- are disjoint subsets of W_7 lying in the core or its collar with $C \subseteq P^+$, checks that the witness is a nonzero integer vector supported on vertices of one colour, recomputes $\hat{q}_P(\mathbf{y})$ from (5), and checks the strict integer inequality (6) in arbitrary precision integer arithmetic;
- (iii) checks that every node of the cover stores a listed pattern consistent with its subcube, that its out-neighbours carry exactly the first-failure children of the node, and that every rooted core has a root node with subcube $(C_0, \emptyset)$; and
- (iv) generates the 116 polyhexes with at most six cells, verifies the counts (8), and decides positive semidefiniteness of the two colour blocks of each exactly over $\mathbb{Q}(\sqrt{5})$, by the symmetric Gaussian elimination, pivoting on positive diagonal entries, described in Section 6, reporting the survivors. This is the cross-check of Section 6; the witnesses of (9) are checked separately, as described below.

The run takes under two seconds and reports the three survivors of Proposition 6.1. Arbitrary precision integers are supplied by the Boost multiprecision library or, equivalently, by GMP; the program has been compiled and run with both, with identical output. Machine integers would in fact suffice. If the support of $\mathbf{y}$ has size s and its entries are at most b in absolute value, then at most $3s$ white vertices of the tiling meet the support and each of them contributes at most $(3b)^2$ to the first sum in (5), since every vertex of the tiling has degree three; thus $M = |\hat{q}_P(\mathbf{y})| \leq 27sb^2$ and $N = \mathbf{y}^T \mathbf{y} \leq sb^2$, so $0 < 2M - N \leq 55sb^2$ and both sides of the second inequality in (6) are at most $(55sb^2)^2$. In the certificate every witness has $s \leq 15$ and $b \leq 179$, the largest value of N is 89,954 and the largest value of $2M - N$ is 201,144, so the largest

¹They are available at <https://github.com/kguo-sagecode/golden-benzenoid> and are archived at Zenodo, doi:10.5281/zenodo.22996285.

integer actually compared is $4.05 \cdot 10^{10}$ and no overflow of a 64-bit integer is possible. We keep the arbitrary precision arithmetic as belt and braces. The transcript of the run against GMP is the file `seven_hexagon_verification_gmp_rerun.log`.

Independent re-verification

A second verifier, `seven_hexagon_reverify.py`, written in Python from the specification of Sections 5 and 6 rather than from the C++ source, repeats checks (i) to (iii) and then performs tests of a different kind. It draws a rooted core and a random occupancy pattern of W_7 containing it, routes the occupancy pattern through the cover to the node whose stored pattern it matches, and then computes the block M_C of that pattern's core directly and exactly over $\mathbb{Q}(\sqrt{5})$, without reference to the witness, and confirms that it is not positive semidefinite. Two thousand such end-to-end tests were run on the 22,009-pattern certificate with no disagreement; the transcript is the file `seven_hexagon_reverify_shipped_2000.log`. Separately, for each of the $333 + 1448 + 6572 = 8,353$ polyhexes with seven, eight or nine cells, a connected seven-cell obstruction containing the lexicographically first cell was found by direct exact computation, by the program `seven_hexagon_optimality.py`, which also carries out the exact checks behind Proposition 4.2; the transcript is the file `seven_hexagon_statement_scan_h9.log`. This is the conclusion of Theorem 1.3 in the rooted form in which we have stated it.

The integer witnesses of Section 6 are shipped as the file `small_polyhex_witnesses.txt`, which also records the characteristic polynomials and their factorizations for the three members of the golden gap class. A third verifier, `small_polyhex_verify.py`, regenerates the 116 polyhexes independently, checks the counts (8), and verifies each of the 113 witnesses against (9) in integer arithmetic, recomputing N and M from the adjacency matrix; it then checks the three factorizations and locates the smallest positive root of each exactly. It uses no library beyond the Python standard library. The transcript is the file `small_polyhex_verify.log`.

The exact checks of Proposition 4.2 were carried out in the same exact arithmetic, and the numerical values quoted in this paper were computed in double precision as a cross-check.

What the computation does not establish

The certificate, and not the search that produced it, is the object of record. Nothing in this paper depends on how the cores, the patterns or the cover were found: the verifier of Section 7 rebuilds W_6 and W_7 , re-enumerates the rooted cores, recomputes every $\hat{q}_P(\mathbf{y})$ and every integer inequality, and re-derives the first-failure partition at every node, so a search that was mistaken, or adversarial, could not produce data that passes. We therefore make no claim about the search, and none is needed.

The computation establishes that the data satisfies the specification of Sections 5.1 to 5.3. It does not, and cannot, check the specification itself, which consists of five facts: that a polyhex rooted at its lexicographically first cell lies in Λ^+ , which is (2) and is immediate from the definition of the lexicographic order; that the collar of a rooted core lies in W_7 , which is Lemma 5.1; that $\hat{q}_P$ bounds the quadratic form, which is Lemma 5.2; that the integer inequality (6) gives $\mathbf{y}^\top M_C(H) \mathbf{y} < 0$ for every rooted polyhex H matching the pattern, which is Lemma 5.3; and that the first-failure children partition a subcube, which is Lemma 5.6. These proofs are short and are given in full above. The second verifier is independent code but was written from the same specification, so on its own it detects errors in the data rather than in the specification. The end-to-end tests and the scan of polyhexes with at most nine cells are the checks that bypass the specification, since they compare the conclusion of Theorem 1.3 against exact spectra computed directly.

Declaration on the use of AI tools

The strategy of the proof is the author's: to prove the stronger statement, Theorem 1.1, rather than Corollary 1.2, by exhibiting a local block of $A^2 - \vartheta^2 I$ that is not positive semidefinite, following the approach of Guo and Royle [10]. The computations showing that six hexagons do not suffice (Section 4.2) were carried out by the author with the assistance of Claude (Anthropic).

The argument for seven hexagons in Section 5 is not the author's. The author's plan for this step rested on a much larger case analysis. A draft of the paper containing that plan and a proof sketch of a much more computationally-dependent proof with more case analysis was given to ChatGPT 5.6 Pro (OpenAI), which completed it with a different argument, including Lemmas 5.1, 5.2, 5.3 and 5.6, none of which the author had drafted. This argument is shorter than the author's original one and a forthcoming edition will reconcile the two approaches to give a more intuitive proof. Claude was also used in editing the text of the manuscript.

A Specification of the certificate

This appendix says what the two files contain and how they are to be read, what a verifier must recompute, and how a certificate may be searched for. It is meant to be enough to rebuild all of it from nothing. We stress at the outset that the search needs no correctness argument of any kind: whatever produces the files, the checks of Section 7 are applied to them afterwards, and data that is wrong in any respect fails those checks.

A.1 Conventions

Cells carry axial coordinates as in Section 2. We place the cell (q, r) with its centre at the point $(2q + r, 3r)$ of the plane, so that its six vertices are that centre translated by

$$(1, 1), \quad (0, 2), \quad (-1, 1), \quad (-1, -1), \quad (0, -2), \quad (1, -1),$$

listed in cyclic order, and the six edges of the cell join consecutive vertices in that list. Every vertex of the tiling then has integer coordinates (x, y) with $y \not\equiv 0 \pmod{3}$, two cells are adjacent exactly when they share two vertices, and the colour of a vertex is $y \pmod{3}$, which takes the values 1 and 2 and is constant on each colour class. Each edge of the tiling lies on exactly two cells, which is the fact the verifier uses to decide whether an edge is present.

A.2 The atlas file

The atlas is a stream of whitespace-separated tokens. It opens with the magic word **SEVEN_HEXAGON_OBSTRUCTIONS_V1** and then has three counted sections. The first is the token **W7**, the number 85, and then 85 records **W j q r** giving the cells of W_7 ; the index j runs from 0 and fixes the numbering of W_7 used everywhere else. The second is the token **CORES**, the number 3,652, and then that many records **C j** followed by the seven coordinate pairs of the j th rooted core. The third is the token **PATTERNS**, the number 22,009, and then that many pattern records. A pattern record is

$$\begin{aligned} & \mathbf{P} \quad j \quad c \quad k_+ \quad b_1 \cdots b_{k_+} \quad k_- \quad b'_1 \cdots b'_{k_-} \\ & \quad k \quad x_1 \quad y_1 \quad c_1 \quad \cdots \quad x_k \quad y_k \quad c_k, \end{aligned}$$

where j is the pattern's index, c the index of its core, the b_i are the W_7 -indices of the cells of P^+ , the b'_i those of P^- , and the triples give the witness: $\mathbf{y}(x_i, y_i) = c_i$, and $\mathbf{y}$ vanishes elsewhere. Note

that a coefficient c_i is permitted to be zero, so the number k of recorded entries is an upper bound for the size of the support and need not equal it.

A.3 The cover file

All integers in the cover are unsigned and little-endian. The file begins with the eight bytes 7HEXCVR1, followed by three 32-bit integers: the number of nodes, the number of rooted cores and the number of patterns. The node records follow, one for each node in order of its index. A node record is a 32-bit integer giving the index of the pattern stored at the node, a 32-bit integer giving the number m of arcs leaving it, and then m arc records of six bytes each. An arc record is one byte giving the index in W_7 of the cell of the corresponding literal, one byte giving the value that the literal asserts, namely 1 for present and 0 for absent, and a 32-bit integer giving the index of the node carrying the first-failure child in which that literal is violated. The arcs of a node are listed in the fixed order of the literals of its pattern that are undecided by its subcube: those asserting presence first and then those asserting absence, each group in increasing order of the index in W_7 . A leaf has $m = 0$. The file ends with one 32-bit integer for each of the 3,652 rooted cores, in the order in which the cores are listed in the atlas, giving the index of the root node for that core. There are no trailing bytes, and a verifier should check that.

A.4 Recomputing the bound

The one quantity a verifier must get exactly right is $\hat{q}_P(\mathbf{y})$, and it is determined by the pattern alone. It is defined by (5); we repeat here only what must be got right about the bookkeeping, and refer to Section 5.2 for the classification of edges itself.

Two conventions govern everything. First, only the admissible collar matters: a cell of the collar of C that lies outside Λ^+ is absent from every rooted polyhex and is to be treated as forced absent, never as undetermined. Equivalently, and this is the form to implement, build $\text{collar}^+(C)$ rather than the full collar, record for each cell of it its index in W_7 , and let a cell carry no weight at all if it has no such index. Getting this wrong is the one error that silently inflates $\hat{q}_P(\mathbf{y})$ and rejects a valid certificate. Second, the cells of C itself are forced present, whether or not they are also listed in P^+ .

Concretely: build $\text{collar}^+(C)$, and record for each edge e of the tiling carried by C or by $\text{collar}^+(C)$ a pair consisting of a flag saying whether some cell of C carries e , and the set of W_7 -indices of the cells of $\text{collar}^+(C)$ that carry e . An edge with an empty record is discarded. Now take each vertex v in the support of $\mathbf{y}$ and each edge vw at v , classify it as forced, undetermined or discarded as in Section 5.2, group the edges by w , and form s_w and the $t_{w,j}$. Then $\hat{q}_P(\mathbf{y})$ is given by (5). Since w has three edges in the tiling, $k_w \leq 3$ and the maximum over J may be taken by inspecting at most eight subsets; alternatively, and this is worth knowing for the search,

$$\max_J \left(s_w + \sum_{j \in J} t_{w,j} \right)^2 = \max \left\{ \left(s_w + \sum_{t_{w,j} > 0} t_{w,j} \right)^2, \left(s_w + \sum_{t_{w,j} < 0} t_{w,j} \right)^2 \right\}, \quad (10)$$

because $J \mapsto s_w + \sum_{j \in J} t_{w,j}$ is affine in the indicator of J and so attains its largest and smallest values by taking all the positive $t_{w,j}$ and all the negative ones respectively.

A.5 Checking the cover

A verifier walks the cover from each root. At a node it recomputes, from the node's subcube and the literals of the stored pattern, which literals are undecided; it checks that none is violated by

the whole subcube, so that the pattern is consistent with it; and it checks that the arcs leaving the node are exactly those literals, in the prescribed order, each leading to a node whose subcube is the one obtained by deciding the literals up to and including that one, with the last decided the wrong way. A node reached twice must carry the same subcube both times, and a verifier should check that as well, since the cover is a directed acyclic graph with sharing and not a tree. Nothing else about the shape of the cover matters.

A.6 Finding a certificate

We describe one method that works. It is a heuristic and we make no claim for it beyond that; its output is checked afterwards.

Fix a subcube, that is, a set of cells of W_7 declared present and a disjoint set declared absent. For a candidate core C and a colour class, (10) turns the quantity we need to make negative into

$$\widehat{q}_P(\mathbf{y}) + \varphi \mathbf{y}^\top \mathbf{y} = \sum_w \max\{L_w^+(\mathbf{y})^2, L_w^-(\mathbf{y})^2\} - \vartheta^2 \mathbf{y}^\top \mathbf{y},$$

using $2 - \varphi = \vartheta^2$, where L_w^+ and L_w^- are the two linear forms obtained by adjoining to s_w all the positive, respectively all the negative, undetermined coefficients at w . The right side is a convex piecewise quadratic form minus a multiple of the squared norm, and it can be attacked by alternation. Choose a starting vector. At the current $\mathbf{y}$, select at each w whichever of the two subsets attains the maximum, which fixes one linear form L_w per vertex w ; assemble the matrix M whose rows are those forms; and replace $\mathbf{y}$ by an eigenvector of $M^\top M - \vartheta^2 I$ for its least eigenvalue. Repeat until the selection stops changing. Then round: for a range of scale factors λ , take the integer vector nearest to $\lambda \mathbf{y}$ and test it exactly, by the integer comparison (6). Keep the first integer vector that passes, preferring small ones.

If no witness is found for the subcube, some cells of the collar must be decided. Extend the pattern by one or more literals, choosing cells whose status most affects the forms $L_w^\pm$, and try again; the added literals are what the cover then branches on. The core of the pattern need not be the core one started from, and for some subcubes it must not be, as Remark 5.7 records; a search that stalls on a subcube should therefore also try other rooted cores whose cells the subcube permits. Finally, assemble the cover: give each rooted core a root node whose subcube declares the seven cells of the core present and leaves the other 78 positions of W_7 undecided, attach to each node the first-failure children of its pattern, and recurse, sharing nodes that carry identical subcubes. The recursion terminates because each step decides one further cell.

We note two things a reader who repeats this will find. The first is that how often a core needs no literals at all is very sensitive to the search. In the certificate deposited here, 181 of the 3,652 root nodes are leaves, and 342 of the 22,009 patterns carry no literal beyond the seven cells of their core; in a fresh reconstruction of the search we obtained a witness with no literals for about two rooted cores in five. The second is that the witnesses are not canonical: they depend on the starting vector, on the range of scale factors and on the order in which literals are added, so a second search will produce a different atlas and a different cover of different sizes. It is for this reason that we report the certificate we verified rather than any property of the search that found it.

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